

Using Experiments to study Mental Models

Andrea Guido* Danièle Fares†

July 7, 2026

Abstract

Decision-makers in organizations rely on causal representations of how their actions translate into outcomes — mental models — to navigate complex policy environments. Understanding these models is central to organizational cognition, yet observational methods can document what models people hold without establishing how they form, which model is selected when several are consistent with the same evidence, or what shifts them. Experiments fill this gap: by controlling the information environment and varying causal structures, they allow researchers to identify which models are activated, why some are adopted over others, and how they respond to persuasion. This chapter reviews the experimental evidence along these three dimensions. On measurement, we distinguish decontextualized designs — which benchmark model formation against a known data-generating process and reveal systematic failures and narrative-induced distortions — from contextualized approaches that elicit full subjective causal maps through open-ended questions. On selection, the evidence shows that when multiple interpretations are consistent with the available data, individuals typically adopt a single model, and the choice is shaped by optimism, cost-aversion, and cognitive simplicity rather than statistical fit. On persuasion, causal narratives shift models by reorganizing how shared facts are causally connected, without conveying new information. Together, these findings suggest that organizational interventions targeting behavior directly may fail if they leave the underlying causal representation unchanged.

Keywords: mental models, organizational cognition, experiments, causal reasoning, narratives

*Paris School of Business, Paris, France

†University of Bergamo, Bergamo, Italy

1 Introduction

Most economic and managerial decisions are difficult and require a substantial quantity of information to process. Decision makers in organizations are often assumed to be “information workers” (McCall and Kaplan, 1985), that is they “spend their time absorbing, processing, and disseminating information about issues, opportunities and problems” (Walsh (1995)). Such description, however, does not emphasize the fact that individuals operate under bounded rationality (Simon, 1955), which limits the extent to which they make use of available information, even when freely accessible.¹ Behavioral and cognitive sciences (Johnson-Laird, 1983; Denzau et al., 2000) commonly assume that people handle information by first constructing an internal representation of the problem — i.e., a *mental model* — that identifies which features matter and how they relate causally to one another. Since their inception, mental models have permeated organizational research. Scholars have examined how managers and employees form interpretations of complex situations and how these interpretations, in turn, shape organizational decisions (Hodgkinson and Healey, 2008; Walsh, 1995; Secchi et al., 2022; Felin and Zenger, 2017).

However, while interest in organizational cognition has grown steadily over the past decades, the empirical study of mental models has faced a fundamental methodological constraint: observational methods and self-report can document what models people hold, but they cannot establish the causal mechanisms through which models form, how competing models are selected, and under what conditions they change. Experiments fill this gap. By controlling the information environment, varying causal structures, and eliciting explicit reasoning, they allow researchers to identify which mental models are activated, why some are adopted over others, and how they can be used to persuade others. This chapter discusses the recent experimental literature on mental models in economics. Following a conceptual framework

¹Examples of bounded rationality are limited attention (Gabaix, 2014, 2019; Gabaix and Laibson, 2006; Golman et al., 2017; Sims, 2010), and bounded memory (Graeber et al., 2024; Kahana, 2012). These cognitive constraints render decision making a rather complex process, oftentimes leading to sub-optimal outcomes.

centered on mental models in organizational decisions, we review how experiments can shed light on their measurement, selection, and persuasive effect.

The chapter is organized as follows. Section 2 introduces the conceptual framework. Section 3 reviews experimental approaches to studying mental models along three dimensions: how to measure them (Section 3.1), how they are selected when multiple models are consistent with the data (Section 3.2), and how they can be shifted through persuasion (Section 3.3). Section 4 concludes.

2 Conceptual Framework

2.1 Decisional Problem

We begin by outlining a simplified conceptual framework that draws inspiration from Enke (2024), and combines it with recent models of mental model formation (Spiegler, 2016). Its aim is far from being exhaustive and complete but it helps the reader navigate the content of this chapter. We assume a common problem in organization, that of deciding which policy to enact. Let us assume a decision maker (DM) i who has to decide what kind of policy a to undertake in their organization. More formally, they are tasked to maximize the organization welfare, comprised of several dimensions k :

$$\max_a U = \sum_k u(y_k) = \sum_k u(f_k(a, \mathbf{z}))$$

where $u(\cdot)$ is the utility function, and y_k is the outcome for the dimension k which can be considered as the result, f_k , of the decision a and a set of exogenously given economic (e.g., intra and inter-organizational) factors $\mathbf{z}$ – with elements z_j . The decision maker is then interested in choosing the optimal policy a^* that maximizes the organization’s utility.

Example 1: Remote work policy. Consider a manager who must choose a remote work policy $a \in \{\text{full in-office, hybrid, full remote}\}$ for their organization. She cares about two output dimensions: $y_1 = \text{team productivity}$ and $y_2 = \text{employee retention}$, so that:

$$\max_a U = u(f_1(a, z)) + u(f_2(a, z))$$

The vector of relevant factors z includes factors that can interact with the effectiveness of the policy (e.g., commute distances, job market tightness, task interdependence, and home office quality) and mediators of the policy effect on productivity (e.g., workers' focus, knowledge sharing, flexibility, social bonds).

Example 2: Performance pay vs. fixed salary. Consider a manager who must choose a compensation scheme $a \in \{\text{fixed wage, performance-based pay}\}$ for their team. She cares about three output dimensions: $y_1 = \text{individual effort and output}$, $y_2 = \text{cooperation and knowledge-sharing among workers}$, and $y_3 = \text{worker wellbeing}$, so that:

$$\max_a U = u_1(f_1(a, z)) + u_2(f_2(a, z)) + u_3(f_3(a, z))$$

The vector of relevant factors z includes workers' social preferences, intrinsic motivation, peer effects, and the degree of task complementarity within the team.

2.2 Mental Models

How do people form f ? An important aspect to notice is that the link between policy actions and their consequences is not always unique among decision makers. They can hold different views and mental representations of reality, which lead to distinct optimal policies.² In the remote work example, one may assume that f_1 and f_2 are likely to pull in opposite directions:

²We refer to mental models rather than just models to emphasize the active part that the decision maker plays in forming them.

a full in-office policy may boost team productivity (y_1) but reduce retention (y_2) if workers value flexibility. But at the same time, one may think that the same in-office policy may instead increase retention because it creates social bonds among workers. As a consequence, a^* depends on how the decision maker interprets the mapping between actions and economic conditions with outcomes, i.e., $f(a, z)$. In what follows, we will refer to such mapping as a mental model. Put differently, a mental model is the decision maker’s representation of how policy actions translate into outcomes, given some economic conditions.

Recent theoretical work in economics and cognitive sciences has proposed two ways to represent mental models. The common departing point is the presence of a DM who observes some *signals* that are in relation with an outcome of interest (e.g., historical data firms’ performances and their characteristics). The DM has to infer the causal relations among outcomes of interest based on the data to, in turn, take the best course of action. We present and discuss two formalizations of mental models that have been used in economics.

Bayesian-network Factorization. Sparked by [Spiegler \(2016\)](#), the first strand of work we present in this section formalizes mental models via Direct Acyclic Graphs (DAGs). DAGs were first introduced in the 1980s by Judea Pearl within statistics and computer science ([Pearl, 2009](#)). Only recently, they found their application in economics ([Spiegler, 2016](#); [Eliaz and Spiegler, 2020](#)) and psychology ([Tenenbaum and Griffiths, 2000](#); [Glymour, 2001](#); [Gopnik et al., 2004](#); [Sloman, 2005](#); [Lagnado et al., 2007](#)). A DAG is a graph including a set of factors (i.e., nodes) and directed links. It provides a *causal* representation of the factors the DM believes explain a given phenomenon, and based on its structure, the DM draws conclusions about which action to take.

Example 1. (Continued) Possible mental models that a DM can form in Example 1 can be represented as in Figure 1. A DM with the mental model in Figure 1a would think that enacting a work-from-home policy would improve productivity by enhancing workers’ flexibility and time spent focusing. On the other hand, under the mental model in Figure

Work from Home $\longrightarrow$ Flexibility $\longrightarrow$ Focus $\longrightarrow$ Productivity

a) Mental model 1

Work from Home $\longrightarrow$ Social Bonds $\longrightarrow$ Knowledge Sharing $\longrightarrow$ Productivity

b) Mental model 2

Figure 1: Representation of mental models from Example 1.

1b, the same work-from-home policy would sever social bonds among colleagues, reducing knowledge sharing, resulting in lower productivity.³

Notice that DAGs are parameter-free representations of mental models. Hence, they do not prescribe how the DM makes inferences about the relation between actions and outcomes. To provide an example of how DAGs can impact the inferences drawn from data, consider the framework introduced in Section 2: a DM who has to take an action a with the goal of maximizing an output y . To figure out the impact that a change in a has on y , the agent may draw on some observable data reporting historical realizations of actions and output, a and y , and an auxiliary variable z over time. In other words, the DM has to compute $p(y|a)$. Assuming all variables are binary, a fully rational DM would apply the conditional version of the law of total probability:

$$p(y|a) = \sum_{z=0,1} p(y|a, z) \cdot p(z|a)$$

However, there are different possible beliefs that the agent might have about the causal relationship between these three variables. The DM might believe that a and z are somewhat related, and jointly produce changes in y . In terms of DAGs, $a \rightarrow z \rightarrow y$. Alternatively, the DM might believe that a and z independently affect y without influencing one another, i.e., $a \rightarrow y \leftarrow z$. Based on the DAG a DM has, he draws different conclusions on the effect of action a onto y , even if they observe the same data. In other words, the DAG, and joint

³In this context, exogenous parameters $\mathbf{z}$ can play a moderating role in the impact between any two factors. Following Example 1, general home office quality may further enhance or hamper the effect of flexibility on focus.

distribution of data, determine a conditional probability $p(y|a)$, that can be calculated using the *Bayesian-Network Factorization Formula* (BNFF).

Example 3. If the DAG of the DM is $a \rightarrow z \rightarrow y$, the BNFF prescribes $p(y|a) = \sum_{z=0,1} p(y|z) \cdot p(z|a)$. This is different from the probability derived via the law of total probability.

Likelihood-Based formalization. A related stream of work, the second we present in this section, defines mental models as likelihood functions that map signals with beliefs. This stream of works, initiated by [Schwartzstein and Sunderam \(2021\)](#), considers the problem of a DM who, before making a decision, needs to interpret some piece of relevant data, and to adopt the model that provide the “most convincing” interpretation. A model is a likelihood function that maps the history to posterior beliefs for the DM, in turn leading the receiver to take certain actions. By most convincing, the framework means the model that fits the data the most.

Both formalizations presented in this section lends themselves well for different purposes. The Bayesian-Network formalization of mental models allows to define a fine-grained view of the DM’s perception and understanding of complex phenomena, such as exceptional inflationary periods ([Andre et al., 2026](#)). The likelihood-based formalization instead lends itself well to study “model persuasion”, i.e., that situation in which a sender (e.g., a financial advisor) seeks to convince a receiver (e.g., an investor) to adopt a model of reality that explain some knowledge (e.g., a firm’s past performance) ([Barron and Fries, 2024](#)).

This exciting theoretical discussion opens up many interesting empirical questions. How do people form or select mental models? How do they shape behavior? But also, from a methodological point of view, how can we measure mental models?

3 Using Experiments to Infer Mental Models

3.1 Common Experimental Setups

Past work has measured beliefs about specific relations between factors in isolation, missing the whole belief system. In this section, we discuss the extant methods used to understand decision makers’ mental models. We divide the literature into two main, non-exhaustive, categories based on the setting in which mental models are measures: decontextualized and contextualized settings. Methods and examples in either category have specific objectives and use different elicitation methods. For example, methods in decontextualized settings often serve to understand people’s causal reasoning ability, by varying complexity, using quantitative benchmarks. On the other hand, in specific contexts and domains, past work has resorted to text data to infer mental model structures, rather than benchmarking them to objective realities.

3.1.1 Decontextualized settings

Recent work has studied how people form and adopt predictive models of simple data generating processes (DGPs) in controlled settings such as laboratory experiments. The advantage of controlled settings relative to other settings is that the researcher’s knowledge of the true DGP provides the counterfactual against which behavioral deviations are measured and causally attributed to the mental model the subject holds. These works complement the long experimental literature studying how and why subjects fail at simple statistical inference (i.e., Bayesian inference problems, see [Benjamin, 2019](#)), by providing evidence of how (and how well) people form predictive mental models. In particular, laboratory experiments allow to measure: *prediction accuracy* (i.e., are decisions fully rational or do they deviate from rationality following a particular mental model?), *decision shifts* (does introducing a causal interpretation — a narrative — change behavior in the predicted direction?), and *choice* among competing interpretations (which selection criteria govern model adoption?).

The experimental setup in [Kendall and Charles \(2022\)](#) represents a good testbed for studying individuals’ deviation from rational expectations. Subjects face a stylized decision problem involving three binary variables: a choice (action a), an auxiliary variable z , and a payoff (outcome y). Rather than real-world labels, variables are coded in neutral terms — choices are "BLUE" or "GREEN," the auxiliary is a triangle or circle, the payoff is "HIGH" or "LOW" — to prevent subjects from importing prior domain knowledge. The joint distribution of these three variables is presented as a compact dataset: a small table of rows, each representing one possible combination of values.

Subjects study the dataset and choose a policy — a number between 0 and 1 representing the probability with which they would take action "BLUE." Subjects are explicitly told that (i) their choice is exogenous — it does not affect the auxiliary variable in any way other than through the relationships shown in the data — and (ii) no hidden variables exist. Under these conditions, a rational subject can determine the causal effect of the action on the payoff simply by counting: $p(\text{HIGH} \mid \text{BLUE})$ and $p(\text{HIGH} \mid \text{GREEN})$ are directly readable from the table, and in one of the datasets shown they are both equal to $\frac{1}{2}$. Participants are paid according to a payoff function that rewards choosing the policy that maximizes expected payoff given their beliefs, while imposing a cost for deviating from the center ($a^* = \frac{1}{2}$). A participant with rational expectations (see Section 2.1) who recognizes that action and outcome are independent would choose exactly $\frac{1}{2}$. Any deviation from this choice can be interpreted in the light of some false causal model, for example, a model in which the choice causes the auxiliary variable, or both of them jointly impact the outcome.

Another example of a controlled setting used to infer mental models is that of [Kendall and Oprea \(2024\)](#). Their experiment maps out the cognitive landscape of model formation — identifying which DGPs are hard or easy to model, what makes a rule cognitively demanding, and whether subjects who hold a model can articulate it. The setting is stripped down to its essentials: a binary input sequence and a binary output sequence, connected by a

deterministic algorithm. At each time step, the subject observes a letter — either a or b, drawn randomly — and a corresponding output — either x or y — determined by the algorithm. The algorithm is one of eight possible "finite automata" which are essentially simple, deterministic, yet complex, algorithms. The subject's task is to figure out the rule that generated the data. Four types of algorithms are used, each representing a qualitatively different kind of relationship between inputs and outputs.

Each subject completes eight tasks, one per algorithm. Each task has three parts that collectively operationalize three distinct aspects of mental model formation. In Part 1, participants simply observe input-output pairs generated by the algorithm. There is nothing to do except read — this is the exposure phase, analogous to a person watching how a system behaves before acting on it. In Part 2, participants enter the prediction phase. Twelve new inputs appear in sequence, and before each output is revealed, the subject must predict it. Crucially, no feedback is given during this phase — the subject commits to each guess without knowing whether the previous guess was right. Payment of 0.35\$ per correct prediction gives subjects a financial stake in forming the correct model. A subject who has extracted the algorithm will predict correctly at a high rate; one who has not will be no better than chance. The researcher classifies a subject as having extracted a model if their 12 predictions are internally consistent with some data-compatible algorithm. In Part 3, the subject shifts from implicit behavior to explicit description. They write a free-form verbal explanation of the rule they believe governs the system. The subject is paid based on how well a future participant in the experiment predicts, creating a strong incentive to write a description that is both accurate and comprehensible. A subject who has extracted the algorithm but cannot translate it into words — a plausible situation for rules that are felt more than understood — will fail to articulate it even though their own predictions are correct.

3.1.2 Contextualized settings

Controlled and decontextualized settings lend themselves well to studying deviations from rationality and provide insight into causal reasoning, thanks to the fact that one can control the DGP. However, if a researcher is interested in obtaining insights regarding a specific context, or in the factors deemed to play a role in explaining a phenomenon, and how they are interconnected, recent work has suggested bottom-up approaches to infer mental models. Such approaches rely on open-ended questions to infer “what is on top of people’s minds” ([Haaland et al., 2024](#)).

For example, in [Andre et al. \(2026\)](#) the core elicitation tool is a single open-ended survey question: respondents are asked to explain, in their own words, which factors caused the recent increase in the inflation rate. No pre-specified options are provided — the question is entirely free-form, allowing respondents to mention any combination of causes and connections they see fit. The free-text responses are then hand-coded according to a pre-defined taxonomy into DAGs by trained coders following a two-stage protocol. In stage 1, coders identify the factors mentioned in open-ended answers (e.g., supply chain disruptions, government spending, pandemic) using the taxonomy. These factors represent the nodes of the DAG. In stage 2, coders identify directed edges, that is, the causal connections between factors that the respondent asserts, with inflation as the final outcome node. The result is a DAG for each respondent, capturing both what they think caused inflation and how they think the causes relate to each other.

As main outcomes, this elicitation method allows to elicit the individual-level and aggregate-level mental model, and linking it to behavioral and psychological measures elicited in the same questionnaire. Moreover, from each DAG, one can extract several structural measures: number of factors (nodes), number of edges, longest causal path to final outcome, whether the DAG spans both demand- and supply-side factors, and number of non-terminal links (connections not ending directly at inflation). These serve as continuous measures of mental

models' richness and complexity.

Two methodological choices are worth noting. First, the open-ended format avoids framing effects: respondents are not anchored to a menu of pre-specified factors, so the content of answers is endogenous to respondents rather than imposed by the researcher. This is methodologically central — the goal is to measure the full subjective causal model, including politically loaded or non-expert explanations, that a structured instrument might miss or suppress. Second, representing responses as DAGs allows to capture causal structure, rather than just topic salience. For example, a respondent who mentions government spending as an indirect cause (e.g., $\text{spending} \rightarrow \text{demand} \rightarrow \text{inflation}$) is treated differently from one who treats it as a direct cause.

At the same time, two important practical questions arise when using this elicitation method. The first concerns annotation reliability. Identifying whether a respondent mentions a factor is substantially easier than inferring whether they assert a causal link between two factors. Free text rarely states connections explicitly ("A causes B") — coders must infer direction from phrasing, which introduces ambiguity. In [Andre et al. \(2026\)](#), two independent coders applied the scheme to 200 randomly selected responses. Factor-level agreement reached 88% while connection-level agreement reached 77%. The 11pp gap between factor and connection agreement (88% vs. 77%) may reflect the higher difficulty and arbitrariness in inferring causal links. It implies that the edge structure of the DAGs carries more measurement noise than the node structure. The second question concerns how well human coders compare to Large Language Models (LLMs). With the advent of LLMs capable of reasoning and inferring context from text, researchers may think of deploying them on a large scale to speed up the annotation process. This is becoming common in recent empirical work (e.g., analyzing tweets; see [Gehring and Grigoletto \(2025\)](#)), where authors have started to use LLM-based (e.g., OpenAI) classification via API calls to classify content from text and infer mental models. However, the hardest hurdle remains that of inferring semantic relations

among identified factors.⁴ It is worth noting that these works take as input “raw” data (i.e., text collected from social media or newspapers) whose underlying causal structure is not always clear and easy to infer. There may be ambiguities as per the meaning of words and verbs, or simply the context in which a text has been written. An interesting approach to explore would be that of using LLM-based methods on answers from the methods seen in past experimental work (e.g., [Andre et al., 2026](#)).

3.2 Mental Model Selection

In previous sections, we have covered the experimental methods used to measure how DM’s expectations deviate from rational ones, how a DM forms expectations based on causal representations (e.g., DAGs), and how we can go about measuring mental models bottom up using texts. A natural next question is how individuals choose among multiple possible mental models. In many economically relevant settings, the problem is not simply to update beliefs within a known model, but to decide which model should be used to interpret the available data in the first place. This distinction matters because the same evidence can support different interpretations, and those interpretations can imply different predictions, different causal stories, and thus different actions. Experimental work suggests that this selection process is far from neutral since individuals do not always identify the objectively best-fitting interpretation of the data. Instead, model selection appears to reflect a combination of statistical fit, selective attention to variables, and the attractiveness of a model’s implications for actions. Mental model selection therefore emerges as a distinct source of heterogeneity in judgment and behavior.

In this section, we discuss such interesting, understudied questions by reporting the results from three recent experimental works that deploy setups similar to those described in [Section 3.1](#).

⁴[Lange et al. \(2022\)](#) have suggested a pipeline to overcome this difficulty.

A first way to study mental model selection experimentally is to provide subjects with several candidate models and examine how they weight them after observing data. This is the strategy used by [Aina and Schneider \(2025\)](#). Their starting point is that belief updating is usually studied as if individuals know the single correct way to interpret a signal, whereas in many real settings, the same evidence can be explained by multiple competing models. Their experiment therefore shifts the problems from standard Bayesian updating to updating under model uncertainty. Their experimental design builds on the balls-and-urns task of [Edwards \(1968\)](#), but adds another layer of uncertainty. Participants face two possible states of the world, A or B , and observe a noisy signal in the form of a colored ball drawn from an urn. Unlike in the standard setup, however, subjects are shown two possible data-generating models at the same time, one of which is randomly selected as the true process. Under each model, Bag A and Bag B contain different mixes of purple and orange balls. Participants do not know which model or which bag generated the observed signal; they only see the color of the drawn ball. For each possible model, the experimenter explicitly states what that model would imply about the probability that the true state is A after observing the signal. Participants must then decide how much weight to place on each model’s prediction when forming their own posterior belief.

The main result is that many subjects do not behave like Bayesians who average across models using posterior model probabilities. Instead, they find that the most common behavior is placing full weight on a single model and ignoring the rest. In particular, many participants behave as if they select the model that best fits the signal *ex post*, rather than averaging across the competing interpretations. Some subjects do assign positive weight to both models, in a way closer to Bayesian reasoning, but even in those cases, the weighting is often biased. These distortions have important consequences: they can generate Bayes-inconsistent beliefs, in the sense that subjects become more confident in one state regardless of which signal they observe. The contribution of their study is to show that when individuals face rival explanations of the same evidence, many simplify the problem by adopting a single model as

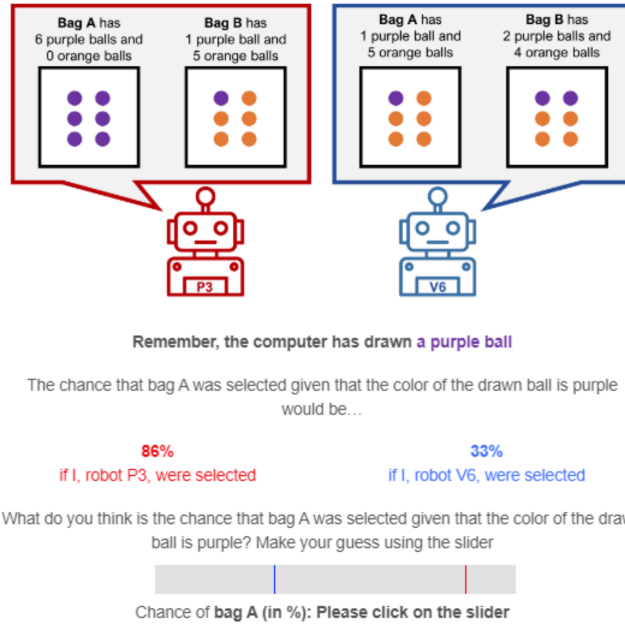


Figure 2: From [Aina and Schneider \(2025\)](#)

an interpretation rather than averaging across alternatives. Mental model selection is thus not a marginal feature of reasoning under uncertainty; it is a central source of bias in how evidence is translated into beliefs.

A second approach does not present subjects with explicit competing models, but instead asks what models they construct for themselves from a set of observations. This is the approach used by [Fr  chette et al. \(2024\)](#). In contrast to [Aina and Schneider \(2025\)](#), where the model space is given exogenously by the experimenter, here, the authors show that mental model selection often begins earlier, at the stage where individuals decide what patterns to abstract from raw data and which variables deserve attention at all. Their question is: when people are shown raw observational data and no theory, what kind of models do they build from it? To answer, they use an experiment with a deliberately simplified setting. Participants observe data from a hypothetical machine with three variables: two lights and a sound. They are first asked to study examples generated by this machine and to take notes. Later, they must predict whether the machine will make a sound, based only on their notes and partial information about the light. This design allows the authors to use both the notes participants

write and the predictions they later make in order to infer the underlying mental models they seem to have formed, in particular, what they think is the relevant statistical relationship among the variables.

The findings show that participants do extract predictive structure from the data, but do so imperfectly and heterogeneously. Average prediction accuracy is 71%, above the random benchmark of 50%, but below the optimal accuracy benchmark of 81%. More importantly, most of the variation in performance does not come from differences in the datasets that participants happen to observe, but from differences across participants themselves. Only 4% of the variance in prediction optimality is attributable to the datasets, while 96% comes from variation across individuals. Subjects shown exactly the same data often disagree about which variables matter and how they matter. Indeed, the authors identify two broad types of error. Some participants systematically ignore variables that are, in fact, relevant for prediction. Others systematically condition on variables that are irrelevant. The main takeaway is that disagreement in judgment is not only a matter of noisy inference or limited effort; it reflects the fact that people build and choose different mental representations of the same environment.

A third line of work shows that mental model selection is shaped not only by how well a model fits the evidence, but also by what the model implies for action. This is the focus of [Ambuehl and Thysen \(2024\)](#), who study how people choose between competing causal interpretations of the same correlational data. Subjects in their experiment must choose a costly action that affects an outcome through a true but unknown data-generating process. They do not observe the true causal structure directly. Instead, they are given correlational data generated by that structure and are shown a menu of two or three possible causal interpretations recommended by advisors, represented by DAGs. Each interpretation comes with an associated action recommendation and promised payoff. Participants must decide which interpretation to follow, and the recommended action is then implemented. This design

thus makes it possible to distinguish whether subjects choose models because they fit the data best, because they imply more attractive outcomes, or because they recommend and imply less costly actions.

The results reveal strong heterogeneity in model choice. Some individuals do attempt to identify the correct interpretation from the data. In one of the samples composed of students, about 37% consistently identify the correct interpretation using the most demanding criterion. However, around 46% appear to use no effective data-based criterion at all. Instead, their choices are guided by the implications of the model. About 17% of student subjects systematically follow the most optimistic interpretation, which is the one promising the highest payoff, while 29% systematically follow the interpretation recommending the least costly action, which the authors interpret as a cautious response to causal uncertainty. In the general population sample, these tendencies are even stronger: only about 6% consistently identify the correct interpretation, while more than 40% follow the most promising model. The broader lesson is that, empirically, people do not select causal models on the basis of statistical fit alone. Mental model selection can also be shaped by optimism, cost avoidance, and preferences over model form. A model may be chosen not because it best explains the data, but because it licenses a more appealing or less demanding course of action.

These three papers point to a common takeaway. Once multiple interpretations of the same evidence are possible, mental model selection becomes a distinct source of heterogeneity in judgment and behavior. It is a substantive cognitive process, one that shapes how data are understood, which variables are taken seriously, and what actions appear justified. For this reason, experiments are particularly valuable: by controlling the model space, the informational environment, and the action consequences attached to each interpretation, they make it possible to distinguish disagreement in beliefs from disagreement in the models through which the evidence is read. They allow researchers to distinguish failures of inference from failures of representation, and epistemic model choice from motivated or action-based

model choice. Recognizing that mental model selection is endogenous to the way evidence is interpreted naturally leads to the next issue: how external interventions, such as narratives or framing devices, can shift the model that the DM adopts.

3.3 Mental Model Persuasion

Having seen that DMs may choose among several plausible interpretations of the same evidence, we now turn to the possibility that this choice can be steered by others. Indeed, once mental model selection is understood as contingent rather than automatic, it becomes important to ask whether others can intervene in this process by making one interpretation more compelling than another.

In many economically and organizationally relevant settings, persuasion does not necessarily operate by changing the underlying facts available to the DM. Rather, it can operate by changing how those facts are interpreted. In other words, persuasion often works at the level of the mental model: it offers a representation of how observed events fit together, which factors should be treated as relevant, and what conclusions should be drawn from them. This influence can take at least two forms. First, individual persuasion, in which a sender can directly shape the model adopted by a receiver. Second, there is a broader social transmission channel through which models spread, are repeated, and become stabilized as individuals exchange interpretations with one another. This is particularly important in organizations, where managers, advisors, or peers often observe broadly similar evidence, yet shape decisions not only by advancing their own interpretations, but also by influencing which interpretations circulate and become collectively accepted.

A useful starting point is the framework developed by [Schwartzstein and Sunderam \(2021\)](#), who formalize what they call model persuasion. In their framework, a sender influences a receiver not by changing the information available but by proposing a model that shapes how existing data is interpreted. The receiver adopts the model that best fits the observed

history given her prior beliefs, and then updates within that model. The central result is "fit-movement tradeoff": models that fit the data well tend to move beliefs less, while models that move beliefs strongly fit the data less well. From there come several implications. First, misleading persuasion can succeed even when the true model is available, because the wrong model may fit the realized data better. Second, persuasion is particularly effective in noisy environments, where the default interpretation struggles to explain what has been observed. Finally, when there is competition between persuaders, it can lead to overfitted models that neutralize the informational content of the data, rather than clarify it. More broadly, their work provides an account of persuasion as model provision: senders influence action not only by recommending what to do, but by shaping the interpretative framework through which evidence is read. Similarly, [Aina \(2023\)](#) develops a related, but more explicitly strategic, framework of persuasion through models. Rather than focusing on a sender proposing a single interpretation of existing data, she studies a sender who can supply a set of models that will shape how future signals are interpreted. After observing a signal, the receiver adopts the model that best fits what she has seen, given her prior beliefs. Persuasion therefore operates through sense-making rather than new information: the sender changes the menu of possible interpretations, not the evidence itself. The study also shows that this form of persuasion depends on prior beliefs: individuals with more balanced priors are more manipulable, while sufficiently different priors may instead become polarized. These two theoretical contributions provide the conceptual foundation for the study of mental model persuasion. They suggest that persuasion is not limited to the communication of new facts or recommendations. It can instead work by supplying interpretive structures that make one reading of the evidence appear more compelling than another. Recent experiments provide direct evidence that DMs are, in fact, influenced in this way.

[Barron and Fries \(2024\)](#) offer the clearest experimental contribution that persuasion can work through interpretation alone. They study a stylized financial-advice setting in which an advisor and an investor observe the same historical company-performance data. The advisor

can then send a message in the form of a narrative about the company’s past and future performance, and the investor combines the observed historical data with this narrative to form a belief about the probability of future success of the company. The authors compare a symmetric-information environment, in which advisor and investor observe the same facts, to an asymmetric-information environment, in which the advisor has an informational advantage. This allows them to separate persuasion through interpretation from persuasion through information transmission. The authors thus study four dimensions of narrative persuasion: are narratives persuasive, what makes them persuasive, how senders construct them, and how the presence of a competing narrative changes sender behavior. Their results show that narratives do shift beliefs, even when the advisor has no private information. They further show that persuasiveness depends on the narrative fit, that is, how well the narrative coheres with the publicly observed facts. Advisors appear to understand this, and thus strategically tailor their narratives to the data. When faced with competing narratives, they respond by constructing messages that fit the evidence better and are less biased. The paper therefore shows that persuasion can be genuinely interpretive: what matters is not only what is known, but how what is known is organized in a coherent explanation. A complementary experimental contribution comes from [Kendall and Charles \(2022\)](#), who show that persuasive narratives don’t need to be deliberately crafted by a sophisticated sender. Their broader analysis studies how causal narratives distort beliefs by changing the way subjects interpret correlations, but one especially relevant result for the present discussion concerns the creation and transmission of narratives. In their ELICIT treatment, subjects first observe abstract datasets and make policy choices, and are then asked to provide free-form advice to future participants. This allows the authors to recover the narratives that subjects themselves generate after trying to make sense of the same data. They find that many subjects spontaneously construct false causal narratives, especially when the data contains correlations that can sustain such interpretations, and that these narratives are much more likely to take the form of Chain stories ($a \rightarrow z \rightarrow y$) than Collider stories ($a \rightarrow y \leftarrow z$). In the

subsequent NATURAL treatment, later subjects receive these subject-generated narratives rather than narratives written by the experimenter. The results show that these "homegrown" narratives are persuasive: endogenously generated Chain narratives shift policy choices away from the policy implied by correct inference, and are often perceived as helpful by receivers, even when they reflect a mistaken causal interpretation of the data. The importance of this result is that it extends the study of persuasion beyond strategic messaging. Narratives can emerge organically from individuals' own mental model construction and still spread in ways that shape the beliefs and decisions of others.

These contributions suggest that mental model persuasion can operate through several related channels. It can take the form of a sender proposing a model that fits existing data and therefore becomes compelling; it can take the form of a sender strategically supplying a menu of models that will later shape interpretation; and it can also arise endogenously, when individuals generate narratives of their own and transmit them to others. Across these approaches, the common theme is that persuasion works not primarily by altering what people know, but by altering how they understand what they know. This matters in collective decision environments, where disagreement often reflects not conflicting facts, but conflicting causal and predictive models through which those facts are interpreted. In such settings, persuasive communication often amounts to the successful transmission of a mental model.

4 Conclusion

Decision makers in organizations operate in complex causal environments: the same policy (e.g., a change in the compensation scheme, a shift to remote work, a reorganization of teams) can produce very different outcomes depending on how agents represent the causal links between actions and outcomes. As this chapter has argued, these representations — mental models — are not merely incidental to behavior but its causal foundation. Experiments are the uniquely suited tool for making mental models visible: by controlling the information

environment, varying causal structures, and eliciting explicit reasoning, they allow researchers to identify which models are chosen, and under what conditions they change.

Two complementary measurement strategies emerge from the experimental literature. De-contextualized designs expose subjects to data generated by a known DGP, benchmarking mental model formation against a ground truth. The results are sobering: formation succeeds only about 40% of the time on average ([Kendall and Oprea, 2024](#)); causal narratives systematically distort decisions in predictable directions ([Kendall and Charles, 2022](#)); and much of what subjects learn is tacit, that is, models that are extracted cannot always be articulated. Contextualized designs, by contrast, elicit the full subjective causal map through open-ended questions and DAG coding ([Andre et al., 2026](#)), linking individual mental models to downstream expectations in ecologically valid settings. The two strategies complement each other: the former provides causal identification of model formation failure; the latter provides ecological validity and domain specificity.

On selection, the experimental evidence converges on two findings. First, when multiple models are consistent with the available data, people do not integrate them probabilistically — they select one, and the selection is often driven by simplicity, optimism, or cost-aversion rather than statistical fit ([Aina and Schneider, 2025](#); [Ambuehl and Thysen, 2024](#)). Second, given identical data, individuals construct systematically different models, with most of the variance attributable to individual-level tendencies rather than the data itself ([Fr chet te et al., 2024](#)). On persuasion, the evidence shows that mental models can be shifted through narratives that reframe how shared facts are causally organized ([Barron and Fries, 2024](#); [Kendall and Charles, 2022](#)). Persuasion works not by providing new evidence but by offering a more coherent causal account — one that connects observed facts into a single interpretive structure. Model change is possible, but it requires engaging causal structure directly.

The evidence reviewed here also speaks to a broader theoretical conversation in organizational cognition and strategy. The theory-based view of [Felin and Zenger \(2017\)](#) argues that great

strategies originate from firm-specific theories of value — contrarian, parsimonious, and generative — and that resources remain inert without an animating theory. That framework, however, leaves open how such theories form, how one theory is selected when multiple interpretations of the same evidence are available, and how they are communicated: indeed, [Felin and Zenger \(2017\)](#) identify the difficulty of transmitting a novel theory to investors and employees as the central governance challenge of the firm. The experimental literature reviewed in this chapter provides empirical microfoundations for each of these open questions — identifying the conditions under which model formation succeeds or fails, the criteria governing model selection, and the mechanisms through which narratives shift the causal representations of others. In this sense, experiments are not only a methodological tool for organizational cognition research; they are the natural complement to theoretical frameworks that treat the economic actor as a theorist.

These findings carry implications for organizational design. If managers and employees systematically hold different mental models of how organizational policies translate into outcomes, interventions that target behavior directly — incentives, mandates, information campaigns — may fail because they leave the underlying causal representation unchanged. More effective interventions will engage the mental model itself: by making causal structure explicit, by providing structured experience with the relevant DGP, or by offering competing models alongside evidence of their relative fit. Designing such interventions requires experiments, not just field observation. More broadly, the experimental toolkit reviewed here has been applied primarily to individual decision settings. The organizational dimension — how mental models are shared and coordinated within teams, how model divergence between hierarchical levels generates conflict, and how organizations learn to revise their causal maps — remains largely unexplored and represents the natural frontier for future work.

References

- Aina, Chiara, “Tailored stories,” *Working Paper*, 2023.
- and Florian Schneider, “Weighting competing models,” *CESifo Working Paper*, 2025.
- Ambuehl, Sandro and Heidi C Thyssen, “Choosing between causal interpretations: An experimental study,” *NHH Dept. of Economics Discussion Paper*, 2024, (07).
- Andre, Peter, Ingar Haaland, Christopher Roth, Mirko Wiederholt, and Johannes Wohlfart, “Narratives about the Macroeconomy,” *Review of Economic Studies*, 2026, p. rdag014.
- Barron, Kai and Tilman Fries, “Narrative persuasion,” Technical Report, WZB Discussion Paper 2024.
- Benjamin, Daniel J, “Errors in probabilistic reasoning and judgment biases,” *Handbook of Behavioral Economics: Applications and Foundations 1*, 2019, 2, 69–186.
- Denzau, Arthur T, Douglass C North et al., “Shared mental models: Ideologies and institutions,” *Elements of reason: Cognition, choice, and the bounds of rationality*, 2000, pp. 23–46.
- Edwards, Ward, “Conservatism in human information processing,” *Formal representation of human judgment*, 1968.
- Eliaz, Kfir and Ran Spiegler, “A Model of Competing Narratives,” *American Economic Review*, 2020, 110 (12), 3786–3816.
- Enke, Benjamin, “The Cognitive Turn in Behavioral Economics,” *Working Paper*, 2024.
- Felin, Teppo and Todd R Zenger, “The theory-based view: Economic actors as theorists,” *Strategy Science*, 2017, 2 (4), 258–271.

- Fréchette, Guillaume R, Emanuel Vespa, and Sevgi Yuksel**, “Extracting models from data sets: An experiment,” *Available at SSRN 5026751*, 2024.
- Gabaix, Xavier**, “A Sparsity-Based Model of Bounded Rationality,” *Quarterly Journal of Economics*, 2014, *129* (4), 1661–1710.
- , “Behavioral Inattention,” in “Handbook of Behavioral Economics: Applications and Foundations,” Vol. 2, Elsevier, 2019, pp. 261–343.
- **and David Laibson**, “Shrouded Attributes, Consumer Myopia, and Information Suppression in Competitive Markets,” *Quarterly Journal of Economics*, 2006, *121* (2), 505–540.
- Gehring, Kai and Matteo Grigoletto**, “Virality: What Makes Narratives Go Viral, and Does it Matter?,” Technical Report 12064, CESifo Working Paper 2025.
- Glymour, Clark**, *The Mind’s Arrows: Bayes Nets and Graphical Causal Models in Psychology*, MIT Press, 2001.
- Golman, Russell, David Hagmann, and George Loewenstein**, “Information Avoidance,” *Journal of Economic Literature*, 2017, *55* (1), 96–135.
- Gopnik, Alison, Clark Glymour, David M. Sobel, Laura E. Schulz, Tamar Kushnir, and David Danks**, “A Theory of Causal Learning in Children: Causal Maps and Bayes Nets,” *Psychological Review*, 2004, *111* (1), 3–32.
- Graeber, Thomas, Christopher Roth, and Florian Zimmermann**, “Stories, Statistics, and Memory,” *Quarterly Journal of Economics*, 2024, *139* (4), 2181–2225.
- Haaland, Ingar, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart**, “Measuring what is top of mind,” Technical Report, ECONtribute Discussion Paper 2024.
- Hodgkinson, Gerard P and Mark P Healey**, “Cognition in organizations,” *Annu. Rev. Psychol.*, 2008, *59* (1), 387–417.

- Johnson-Laird, Philip N.**, *Mental Models*, Cambridge: Cambridge University Press, 1983.
- Kahana, Michael J.**, *Foundations of Human Memory*, Oxford University Press, 2012.
- Kendall, Chad and Ryan Oprea**, “On the complexity of forming mental models,” *Quantitative Economics*, 2024, 15 (1), 175–211.
- Kendall, Chad W and Constantin Charles**, “Causal narratives,” Technical Report, National Bureau of Economic Research 2022.
- Lagnado, David A., Michael R. Waldmann, York Hagmayer, and Steven A. Sloman**, “Beyond Covariation: Cues to Causal Structure,” in Alison Gopnik and Laura Schulz, eds., *Causal Learning: Psychology, Philosophy, and Computation*, Oxford University Press, 2007.
- Lange, Kai-Robin, Matthias Reccius, Tobias Schmidt, Henrik Müller, Michael Roos, and Carsten Jentsch**, “Towards Extracting Collective Economic Narratives from Texts,” Technical Report 963, Ruhr Economic Papers 2022.
- McCall, Morgan W. and Robert E. Kaplan**, *Whatever It Takes: Decision Makers at Work*, Englewood Cliffs, NJ: Prentice-Hall, 1985.
- Pearl, Judea**, *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge University Press, 2009.
- Schwartzstein, Joshua and Adi Sunderam**, “Using models to persuade,” *American Economic Review*, 2021, 111 (1), 276–323.
- Secchi, Davide, Rasmus Gahrn-Andersen, and Stephen J Cowley**, *Organizational cognition: The theory of social organizing*, Taylor & Francis, 2022.
- Simon, Herbert A.**, “A Behavioral Model of Rational Choice,” *Quarterly Journal of Economics*, 1955, 69 (1), 99–118.

- Sims, Christopher A.**, “Rational Inattention and Monetary Economics,” in “Handbook of Monetary Economics,” Vol. 3, Elsevier, 2010, pp. 155–181.
- Sloman, Steven A.**, *Causal Models: How People Think about the World and Its Alternatives*, Oxford University Press, 2005.
- Spiegler, Ran**, “Bayesian networks and boundedly rational expectations,” *Quarterly Journal of Economics*, 2016, *131* (3), 867–897.
- Tenenbaum, Joshua B. and Thomas L. Griffiths**, “Structure Learning in Human Causal Induction,” in “Advances in Neural Information Processing Systems,” Vol. 13 MIT Press 2000.
- Walsh, James P**, “Managerial and organizational cognition: Notes from a trip down memory lane,” *Organization science*, 1995, *6* (3), 280–321.